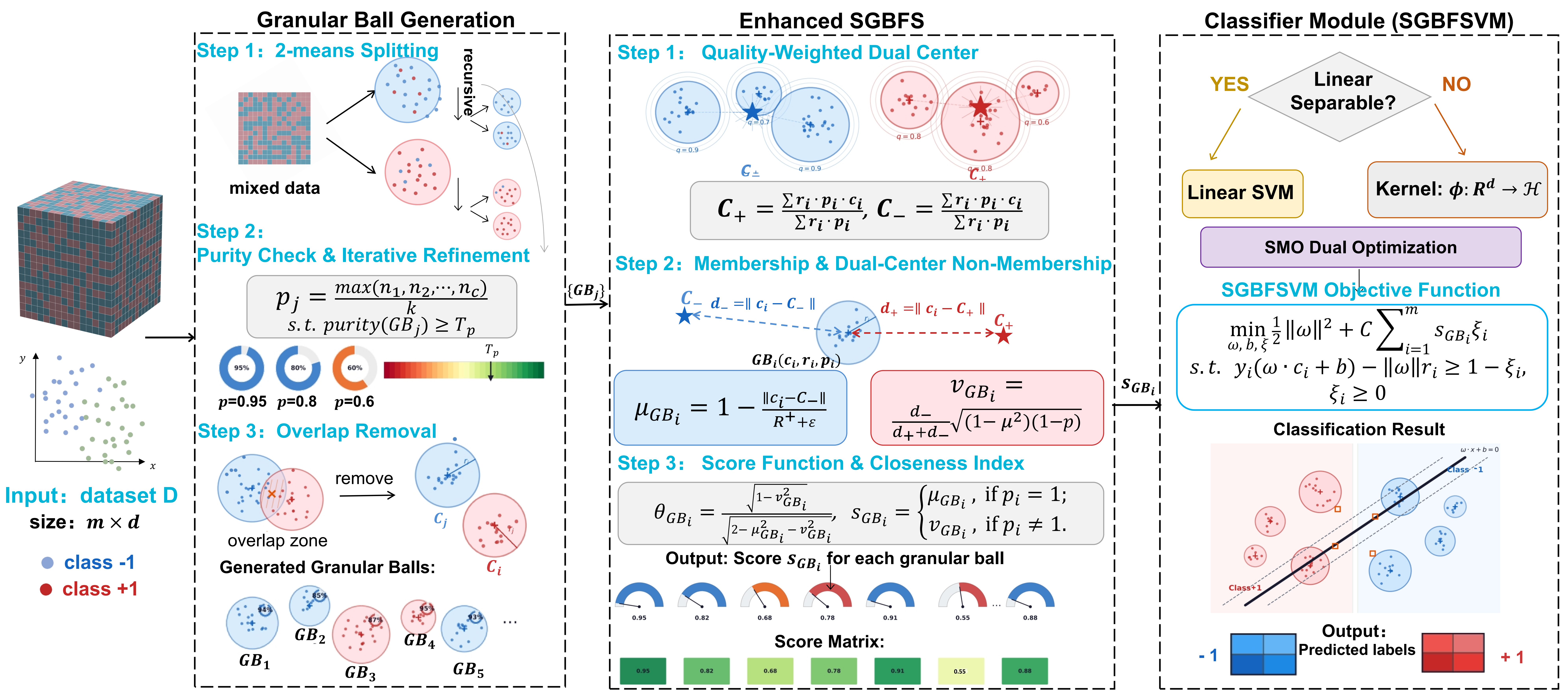

Granular Ball Generation
Step 1: 2-means Splitting
recursive
mixed data
Step 2:
Purity Check & Iterative Refinement
$p_j = \frac{max(n_1, n_2, \cdots, n_c)}{k}$
$s.t.\ purity(GB_j) \geq T_p$
T_p
p=0.95
p=0.8
p=0.6
Step 3: Overlap Removal
remove
overlap zone
C_j
C_i
Generated Granular Balls:
GB_1
GB_2
GB_3
GB_4
GB_5
Input: dataset D
size: $m \times d$
class -1
class +1
$\{GB_j\}$
Enhanced SGBFS
Step 1: Quality-Weighted Dual Center
C_-
C_+
$C_+ = \frac{\sum r_i \cdot p_i \cdot c_i}{\sum r_i \cdot p_i}, C_- = \frac{\sum r_i \cdot p_i \cdot c_i}{\sum r_i \cdot p_i}$
Step 2: Membership & Dual-Center Non-Membership
$d_- = \| c_i - C_- \|$
$d_+ = \| c_i - C_+ \|$
$GB_i(c_i, r_i, p_i)$
$\mu_{GB_i} = 1 - \frac{\|c_i - C_-\|}{R^+ + \varepsilon}$
$v_{GB_i} = \frac{d_-}{d_+ + d_-}\sqrt{(1-\mu^2)(1-p)}$
Step 3: Score Function & Closeness Index
$\theta_{GB_i} = \frac{\sqrt{1-v_{GB_i}^2}}{\sqrt{2-\mu_{GB_i}^2 - v_{GB_i}^2}}, \quad s_{GB_i} = \begin{cases} \mu_{GB_i}, & \text{if } p_i = 1; \\ v_{GB_i}, & \text{if } p_i \neq 1. \end{cases}$
Output: Score s_{GB_i} for each granular ball
0.95
0.82
0.68
0.78
0.91
0.55
0.88
Score Matrix:
s_{GB_i}
Classifier Module (SGBFSVM)
Linear Separable?
YES
NO
Linear SVM
Kernel: $\phi: R^d \rightarrow \mathcal{H}$
SMO Dual Optimization
SGBFSVM Objective Function
$\min_{\omega, b, \xi} \frac{1}{2}\|\omega\|^2 + C\sum_{i=1}^{m} s_{GB_i}\xi_i$
$s.t.\ \ y_i(\omega \cdot c_i + b) - \|\omega\| r_i \geq 1 - \xi_i,$
$\xi_i \geq 0$
Classification Result
Class -1
Class+1
Output:
Predicted labels
- 1
+ 1